

Neuromodulation as a Robot Controller

A Brain-Inspired Strategy for Controlling Autonomous Robots

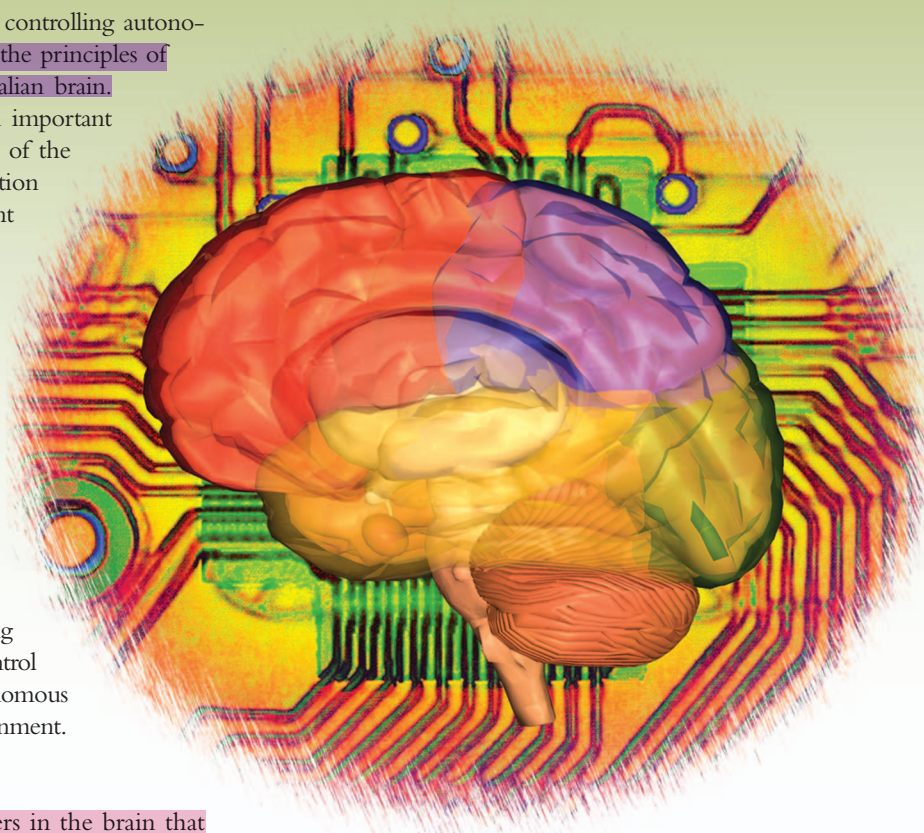
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This article presents a strategy for controlling autonomous robots, which is based on the principles of neuromodulation in the mammalian brain. Neuromodulatory systems signal important environmental events to the rest of the brain causing the organism to focus its attention on the appropriate object, ignore irrelevant distractions, and respond quickly and appropriately to the event [1]. There are separate neuromodulators that alter responses to risks, rewards, novelty, effort, and social cooperation. Moreover, the neuromodulatory systems provide a foundation for cognitive function in higher organisms; attention, emotion, goal-directed behavior, and decision-making derive from the interaction between the neuromodulatory systems and brain areas, such as the amygdala, frontal cortex, and hippocampus. Therefore, understanding neuromodulatory function may provide control and action-selection algorithms for autonomous robots that effectively interact with the environment.

Neuromodulatory Systems

Neuromodulators are chemical transmitters in the brain that can have a strong and lasting effect on an animal's behavior. The neuromodulatory systems include noradrenergic, serotonergic (5-HT), dopaminergic (DA), and cholinergic (ACh) projections from below the cerebral cortex to broad areas of the central nervous system [2]. The origins of these systems are small pools of neurons (of the order of thousands in the rodent and tens of thousands in the human) located below the cortex.

Despite the different origination and chemical signatures of these neuromodulatory systems, there are several commonalities among them.



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- 1) Each of these neuromodulatory systems originates below the cerebral cortex and projects broadly to all regions of the brain.
- 2) Each of these neuromodulatory systems is reciprocally connected with the cognitive areas of the brain such as the amygdala, frontal cortex, and the hippocampus [2].
- 3) The effect of each of these neuromodulatory systems on downstream neuronal targets is similar, i.e., they cause target neural networks to sharpen, resulting in a winner-take-all response [1], [3], [4].

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A computational framework for applying neuromodulatory systems to the control of autonomous robots can be based on the following premises:

- 1) the common effect of the neuromodulatory systems is to drive an organism to be decisive when environmental conditions call for such actions and allow the organism to be more exploratory when there are no pressing events [1], [5]
- 2) the main difference between neuromodulatory systems is the environmental stimuli that activate them. The 5-HT system responds to risks and threats [6], the ACh system sets a level of attentional effort [7], the DA system drives reward anticipation [8], and the noradrenergic system responds to novel and salient objects [9].

From the evidence, it appears that the common effect of the neuromodulatory system is to focus attention on important objects in the environment by increasing the signal-to-noise ratio (SNR) of neuronal responses [1], [5]. Indeed, the major targets of the neuromodulators are areas noted for driving behavior, conditioning responses, focusing attention, and making decisions [2]. The means by which neuromodulatory systems focus an animal's attention is through short bursts of activity in response to important events occurring in its surroundings. During phasic neuromodulation, information from sensory systems (e.g., visual and auditory) is amplified relative to recurrent or associational information [1], [3], [4]. The result of this change in the relative weighting of information is to sharpen responses to environmental input, increase SNR, and drive decisive responses in neural networks. Moreover, neuromodulation gates in learning such that an animal can predict relationships between sensory information and action outcomes [10], [11].

A control system for a robot, which is designed according to principles of the neuromodulatory system, could offer major advantages over conventional systems in carrying out tasks in the face of environmental challenges. Such a system could learn to take appropriate actions depending on context, environmental change, and experience. Neuromodulatory systems drive many of the fundamental behaviors crucial for an organism's survival. Cognitive functions, such as attention, emotion, goal-directed behavior, and decision-making, arise from the interaction between neocortical executive areas and the neuromodulatory systems. Therefore, a controller based on the action of neuromodulation could have much to offer the design of autonomous robots.

In this article, we show, in a neural model, how bursts of ACh, DA, and 5-HT activity can sharpen attention and lead toward appropriate action selection in a cognitive robot. The robot's behavior is guided by a simulation, which has groups of neurons and synaptic connections between these neurons, based on known dynamical and anatomical properties of the neuromodulatory system and its interaction with the surrounding regions of the brain. Although this neurorobot will be used to investigate how neuromodulation can lead to adaptive behavior, principles of this cognitive system may be relevant for the control of robots in general.

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Methods

Robot and Experimental Apparatus

The robot used for the experiments, CARL-1, was constructed in the Cognitive Anteater Robotics Laboratory at the University of California, Irvine [see Figure 1(a)]. It consisted of a two-wheeled mobile base equipped with a charge-coupled device (CCD) video camera having an RF transmitter for vision, IR sensors for obstacle avoidance, and a Wi-Fi device server (<http://www.sena.com>) for communication between the robot and a computer workstation. The pan and tilt position of the camera was controlled by commands to a pair of servomotors. The base of the robot was 10 in in diameter and 8.5 in in height. A distributed network of PIC-18F2680 microcontrollers, which communicated over a controller area network (CAN) interface, read from CARL-1's sensors, controlled CARL-1's actuators, and communicated wirelessly with a computer workstation that contained the neural simulation. Camera video frames were transmitted wirelessly to the Firewire port of the workstation.

CARL-1's environment consisted of a 10 × 10 ft enclosure that contained eight light panels built into the flooring [see Figure 1(b)]. The color of the panels at the four corners were set to cyan, green, magenta, and red at a given frequency and duration through RS-232 communication from the workstation to electronics controlling the panels. All eight panels had IR transceivers that could communicate position information to CARL-1, when it was on top of the panel.

Neural Architecture

The neural simulation that controlled CARL-1 consisted of a visuomotor area, neuromodulatory systems, action areas, and

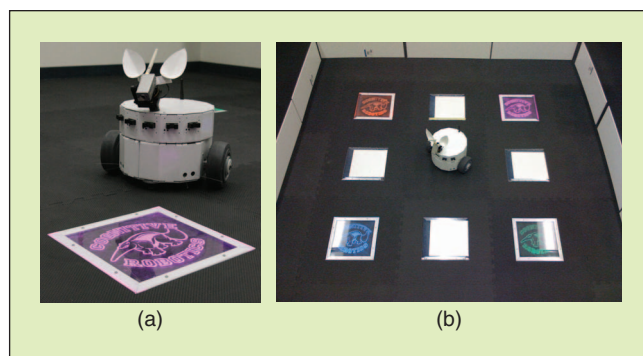


Figure 1. CARL-1 and experimental setup. (a) CARL-1 is a wheeled mobile robot with an RF-CCD camera for vision, IR sensors for obstacle avoidance, and wireless RS-232 for communication with a computer workstation. (b) CARL-1's environment.

Cognitive functions arise from the interaction between neocortical executive areas and the neuromodulatory systems.

behavior drivers (Figure 2). Each ellipse denotes a neural area that contains simulated neurons. The arrows between neural areas denote synaptic projections containing many connections between neurons. Within area connections, inhibitory connections and connections from behavior drivers to action areas are omitted for clarity. The neural simulation contains 6,700 neurons and roughly 1.3 million synaptic connections. The visuomotor area consisted of subareas, each with 15×20 (height $\times$ width) neurons that mapped on CARL-1's field of view. These retinotopically mapped neurons responded preferentially to cyan, green, magenta, and red. The simulated neuromodulatory systems consisted of an ACh basal forebrain (BF) area, 5-HT raphe nucleus (Raphe), and DA ventral tegmental area (VTA). Each of these neuromodulatory areas contained 100 neurons. The action areas consisted of a Find and Flee area that each contained 100 neurons. The behavior driver areas consisted of a good and bad area that each contained 100 neurons.

Neural areas were connected through synaptic projections consisting of probability distributions of connectivity between individual neurons. Neurons in the visuomotor areas had a 10% chance of being connected to neuromodulatory neurons and an initially weak weight (uniformly distributed between 0.05 and 0.10) that could change through experiential plasticity. Within a visuomotor subarea (e.g., red $\rightarrow$ red), the neurons are connected to the neighboring neurons with a two-dimensional (2-D) Gaussian distribution, having a standard deviation of five neurons and an initial weight uniformly distributed between 0.8 and 1.0. Between visuomotor areas (e.g., red $\rightarrow$ green), neurons had a 25% chance of being connected with initial weights uniformly distributed from 0.8 to 1.0 for excitatory connections and from -0.8 to -1.0 for inhibitory connections. The behavior driver neurons had strong all-to-all connections to the neuromodulatory systems and the action areas. Specifically, the

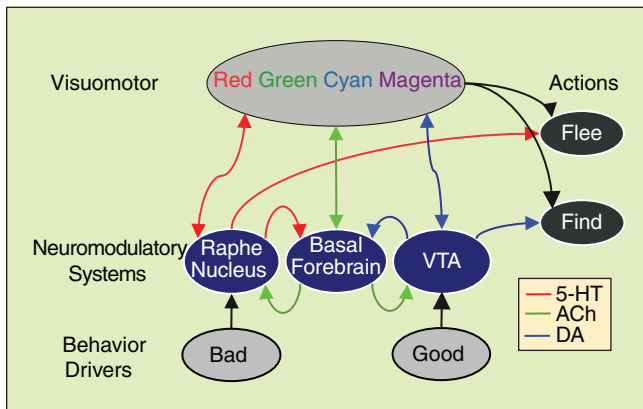


Figure 2. Schematic of neural architecture.

good neurons had excitatory connections to VTA and the Find neurons with weights set to 200, and inhibitory connections to the Raphe and Flee neurons with weights set to -200 . Conversely, the bad neurons had excitatory connections to the Raphe and Flee neurons with weights set to 200, and inhibitory connections to VTA and the Find neurons with weights set to -200 . The VTA neurons projected all-to-all to Find neurons, Raphe neurons projected all-to-all to Flee neurons, and BF neurons projected all-to-all to Raphe and Flee neurons with initial weights uniformly distributed from 0.1 to 0.2.

The visual neurons were set based on input from CARL-1's camera. The OpenCV library (<http://sourceforge.net/projects/opencv/>) was used to subsample the image to 30×40 pixels and run color histogram filters across the image to create separate subareas that responded preferentially to cyan, green, magenta, and red. These responses, which were normalized between zero and one, were used to activate the visuomotor neurons of corresponding colors. These visual responses were connected topographically to the visuomotor neurons with a 2-D Gaussian distribution, having a standard deviation of five neurons and a weight uniformly distributed from 1.0 to 1.5.

Neuronal Dynamics and Synaptic Plasticity

Neural activity in CARL-1 was simulated by a mean firing rate neuron model, where the firing rate of each neuron ranged continuously from zero (quiescent) to one (maximal firing). The activity level of a neuron represented its average firing rate above 100 ms. This model demonstrated the necessary neural dynamics and was efficient enough to run in real time on a robotic platform with sensors and actuators. The equation for the mean firing rate neuron model is

$$s_i(t) = \rho_i s_i(t-1) + (1 - \rho_i) \left(\frac{1}{1 + \exp(-0.1 I_i(t))} \right), \quad (1)$$

where t is the current time step, s_i is the activation level of neuron i , ρ_i is the persistence of the neuron, and I_i is the synaptic input. Visuomotor neurons had a persistence of 0.5, and all other neurons had a persistence of 0.1.

The synaptic input of the neuron is based on presynaptic neural activity, the connection strength of the synapse, and the amount of neuromodulator activity.

$$I_i(t) = \sum_j nm(t-1) w_{ij}(t-1) s_j(t-1), \quad (2)$$

where w_{ij} is the synaptic weight from neuron j to neuron i and nm is the level of neuromodulator at synapse ij .

To simulate the effect of phasic neuromodulation, inhibitory and extrinsic inputs were amplified relative to the overall neuromodulatory activity (i.e., nm is set to be ten times the combined average activity of the simulated BF, Raphe, and VTA neural areas). Connections from the visual input neurons to visuomotor neurons, from neuromodulatory neurons to action neurons, and within the neuromodulatory systems are considered extrinsic. All other excitatory connections are considered intrinsic, and for those connections, nm is always equal to one.

Connections from the visuomotor areas (cyan, green, magenta, and red) to the neuromodulatory areas (BF, Raphe, VTA) and from the visuomotor areas to the action areas (Find and Flee) were subject to synaptic plasticity that depended on the current activity of the presynaptic neuron, the postsynaptic neuron, and the overall activity of the neuromodulatory systems.

$$\Delta w_{ij}(t) = \varepsilon(w_{ij}(0) - w_{ij}(t-1)) + \delta \Theta_{NM}(nm)s_j(t-1)(s_i(t-1) - \Theta_{BCM}), \quad (3)$$

where ε is the decay rate, which is set to 0.00001, that decayed weights back to their original value ($w_{ij}(0)$). This decay acted as a slow forgetting function and prevented over learning. δ is the learning rate set to 0.001, Θ_{NM} is the gating function in which learning occurred only when the level of neuromodulator activity [nm from (2)] was greater than a threshold value, which was set to 2, and Θ_{BCM} was a sliding threshold dictating the amount of synaptic potentiation and depression. The BCM threshold changed as a function of postsynaptic neural activity [12]:

$$\Delta \Theta_{BCM} = 0.001(s_i(t)^2 - \Theta_{BCM}). \quad (4)$$

Action Selection and Behavior

CARL-1's behavior switched between three states: random exploration, orienting and approaching objects of interest (Find), and moving away from noxious objects (Flee). By default, CARL-1 explored unless the difference between the average activity of the Find and Flee neural areas was greater than a threshold of 0.75, in which case the more active area would elicit the corresponding behavior.

During the exploration behavior, CARL-1 would move at a constant speed, while panning its camera to the left and right. CARL-1's turning rate was proportional to the camera pan position, i.e., the further the camera was panned from the midline, the higher the turning rate in that direction.

During Find and Flee behavior, CARL-1 would saccade its camera to the centroid of the most salient object within its field of vision. The most salient object was chosen by applying a Softmax function to the activity of the four visuomotor areas (cyan, green, magenta, and red):

$$P_c = \frac{\exp(5a_c)}{\sum_{i=1}^4 \exp(5a_i)}, \quad (5)$$

where P_c is the probability of choosing color c , a_c is the average activity of visuomotor area c , and a_i is the average activity of visuomotor area i . The Softmax function was applied every time step when CARL-1 was in Find or Flee behavior. Because there were always visual stimuli in its field of view, CARL-1 would inevitably find some object in the environment to point its camera during Find and Flee behaviors. The camera's pan and tilt position was set to the centroid of activity for the chosen color area.

During Find behavior, CARL-1 would saccade its camera to the centroid of the most salient object within its field of vision and orient toward that object. CARL-1's wheel velocity was proportional to the camera tilt position, i.e., the lower the

tilt position the slower the forward velocity. CARL-1's wheel turning rate was proportional to the camera pan position causing it to orient toward the visual target (e.g., if the camera was panned left, the wheel commands turned CARL-1 to the left). This had the behavioral effect of first fixing CARL-1's gaze on a target of interest, followed by turning the body toward the target, approaching the target, and then slowing down when close to the target.

During Flee behavior, CARL-1 would saccade its camera to the centroid of the most salient object within its field of vision but move away from that object. The camera's pan and tilt position, which was set to the centroid of activity for the chosen color area, was used to calculate wheel commands that were, in essence, the opposite of the Find motor commands. CARL-1's turning rate was proportional to the camera pan position but in the opposite direction of the target (e.g., if the camera was panned left, the wheel commands turned CARL-1 to the right), and its velocity was inversely proportional to the camera tilt position, i.e., the lower the tilt position the faster the reverse velocity. This had the behavioral effect of first fixing the gaze on a target of interest, stopping forward progress, and then turning and backing away from the target.

Simulation Computation

The neural simulation contained 6,700 neurons and roughly 1.3 million synaptic connections. The neural simulation was run on a 2xQuad-Core 2.8 GHz Intel Xeon Mac Pro Workstation on the OS X operating system using portable operating system for UNIX (POSIX) threads, OpenCV, and the fast light toolkit (FLTK) graphical user interface (GUI) library. The simulation cycle was fixed at 100 ms. During each simulation cycle, the sensor data were read and processed, the neural activations were calculated, the change in the strength of plastic connections were calculated, behavior was selected, motor commands were sent to CARL-1, and the behavioral and neural data were logged for postexperimental analysis.

Experimental Paradigm

CARL-1 was first trained to associate the color green with Find behavior and red with Flee behavior and then tested under various conditions.

In the training period, CARL-1 explored the environment. Occasionally, when CARL-1 was near a color panel, the operator would press a button on the GUI that would either maximally activate the good area (see Figure 2) and turn the light panel to green or maximally activate the bad area (see Figure 2) and turn the light panel to red. The button would be turned off after several seconds. Training would continue in this manner until CARL-1 had experienced ten good and ten bad events.

In the testing period, CARL-1 explored its environment for 7,500 simulation cycles. The four light panels were set such that each light panel had a different color. Every 8–10 s, the location of the four colors was changed randomly.

Results

Training and testing were repeated with ten different CARL-1 subjects. Each subject consisted of the same physical device but

possessed a unique simulated nervous system differing at the level of synaptic connections. These differences among subjects were a consequence of random draws from probability distributions of connectivity and the variation of initial connection strengths between those neurons. However, the overall neural architecture was similar across all subjects.

Behavioral Results

After training, all ten subjects responded to green stimuli with Find behavior and to red stimuli with Flee behavior (see Figures 3 and 4). During Find behavior, CARL-1 would begin its approach to the green-light panel from several feet away [see Figure 3(a)]. As it neared the light panel, its camera tilted down and it stopped on the panel [see Figure 3(b)]. As soon as the

panel changed to a neutral value color, its camera tilted up and CARL-1 shifted to exploratory behavior [see Figure 3(c)]. During Flee behavior, CARL-1 would stop its approach when it saw the red-light panel from several feet away [see Figure 3(h)]. With its camera centered on the red panel, CARL-1 turned away from the threatening stimulus [see Figure 3(i)]. After completely turning away from the red panel, CARL-1's gaze moved away from the salient object and it shifted to exploratory behavior [see Figure 3(j)].

These behaviors were driven by phasic bursts of activity from the neuromodulatory systems. Green stimuli caused a phasic response in the VTA neurons resulting in an amplification of the green visuomotor area, a dampening of distracter colors, and a strong increase in the Find activity. For example, in Figure 3(d)–

(g), different neural activities are shown just before and during Find behavior. When the light panel switched from red to green [see Figure 3(d) and (f)], the Raphe and Flee neural areas were still active and in competition with the VTA and Find areas. However, a burst of VTA activity amplified Green and Find activity, causing a suppression of Raphe, Red, and Flee activity [see Figure 3(e) and (g)]. In Figure 3(k)–(n), neural activities are shown prior to and during Flee behavior. Just before Flee behavior, there was moderate activity throughout the neural simulation [see Figure 3(k) and (m)]. The Red area has slightly elevated activity, but it was not much more active than other color areas. Moments later, a burst of Raphe activity amplified Red and Flee neuronal responses, and caused a suppression neural activity in other visuomotor, neuromodulatory, and action areas [see Figure 3(l) and (n)].

To further test the necessity of phasic responses in the neuromodulatory systems to generate appropriate behavioral responses, we conducted simulated lesion experiments in all the subjects. In one set of experiments, the activity of neurons in the Raphe area was set to zero, and in another set of experiments, the activity of neurons in the VTA area was set to zero. These lesion groups were compared with a control group that had a complete neural simulation. Lesions of the VTA significantly reduced the number of Find responses [$P < 0.0005$, Wilcoxon rank sum test; see Figure 4(a)] but not the Flee responses [see Figure 4(b)]. On each box in the plot, the central mark is the median, the edges of the box are the 25th and 75th percentiles, the whiskers extend to the most extreme data points

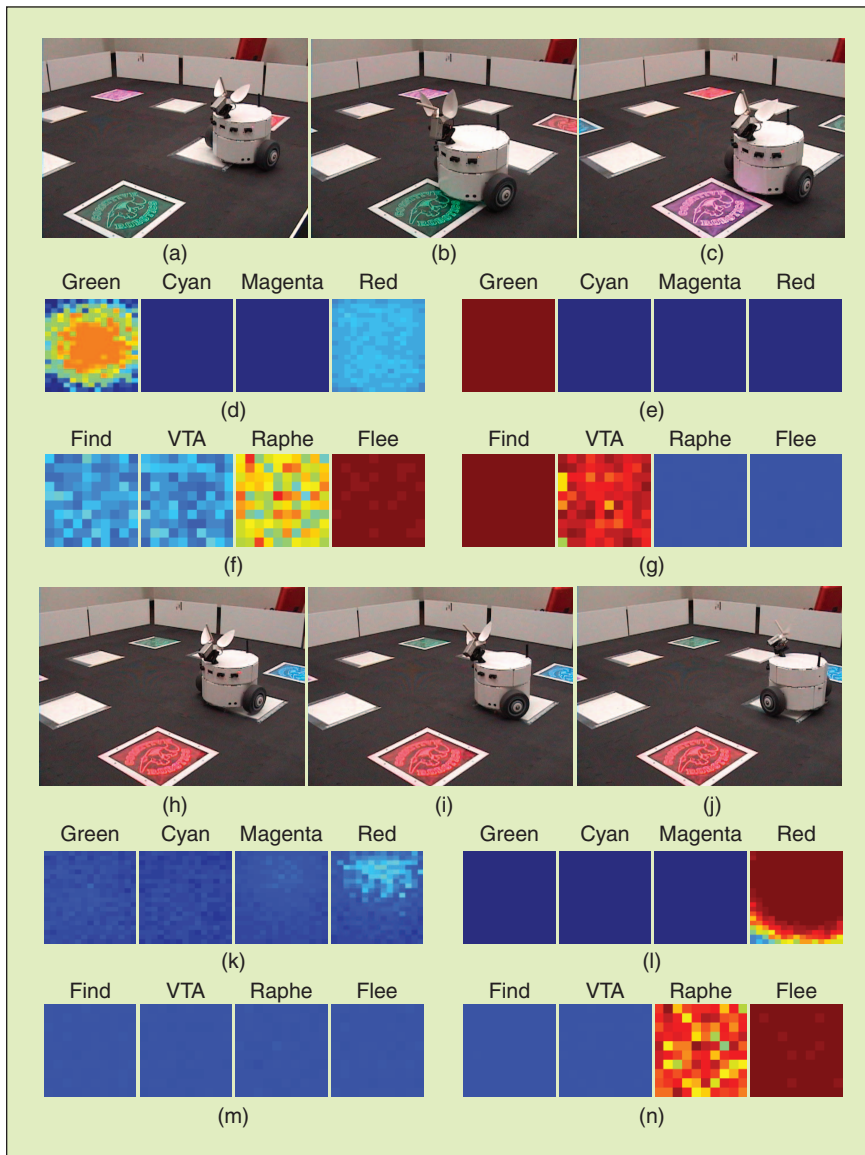


Figure 3. CARL-1 behavior and neural activity. (a)–(g) Find behavior. (a)–(c) Snapshots of CARL-1 during the Find behavior. (d)–(g) The selected neural areas just before and during the Find behavior. Each pixel denotes a neuron, where the activity is color coded from quiescent (dark blue) to maximally active (bright red). (h)–(n) Flee behavior. (h)–(j) Snapshots of CARL-1 during Flee behavior. (k)–(n) The selected neural areas just before and during the Flee behavior.

not considered outliers, and outliers are plotted individually with plus signs. Lesions of the Raphe significantly reduced the number of Flee responses [$P < 0.0005$, Wilcoxon rank sum test; see Figure 4(b)] but not the Find responses [see Figure 4(a)].

The BF is thought to increase attentional effort in challenging conditions. Therefore, we lesioned the BF alone and in conjunction with lesions of other neuromodulatory areas to better understand its functional role. Lesions of the BF area alone did not have a significant effect on behavior (see BF in Figure 4). However, a lesion of BF and VTA completely abolished the Find behaviors [see BF + VTA in Figure 4(a)] and a lesion of BF and Raphe completely abolished the Flee behaviors [see BF + Raphe in Figure 4(b)].

Effect of Phasic Neuromodulatory Responses on Neuronal Activity

Phasic neuromodulatory activity is thought to increase the SNR in neural circuits such that the organism increases the discrimination between salient and nonsalient stimuli. To test this idea, we calculated a SNR metric based on the visuomotor area's response to a target color divided by the visuomotor area's response to all other colors during Find and Flee behavior.

$$SNR = \frac{vis_{tgt}}{\sum_{i=1}^4 vis_i}, \quad (6)$$

where vis_{tgt} is the average activity of the green area during a Find behavior and the average activity of red during a Flee behavior and vis_i is the average activity of visuomotor area i .

Lesions of neuromodulatory responses significantly lowered the SNR in the visuomotor area during behavioral responses. The SNR was significantly lower in the group with VTA lesions than in the control group during Find behavior [$P \ll 0.0001$, t -test; see Figure 5(a)]. A lesion of BF and VTA further reduced the SNR for Find responses [$P \ll 0.0001$, t -test comparing VTA lesion to BF + VTA lesion; see Figure 5(a)]. The SNR was significantly lower in the group with Raphe lesions than that in the control group for Flee behavior [$P \ll 0.0001$, t -test; see Figure 5(b)]. Lesions of both BF and Raphe further reduced the SNR for Find responses [$P \ll 0.0001$, t -test comparing Raphe lesion to BF + Raphe lesion; see Figure 5(b)]. Plots show the mean SNR for the ten subjects with an intact simulated nervous system (control), lesion of the simulated Raphe nucleus (Raphe), lesion of the simulated VTA, lesion of the BF, and lesions of multiple areas (BF + Raphe and BF + VTA). Error bars denote the standard

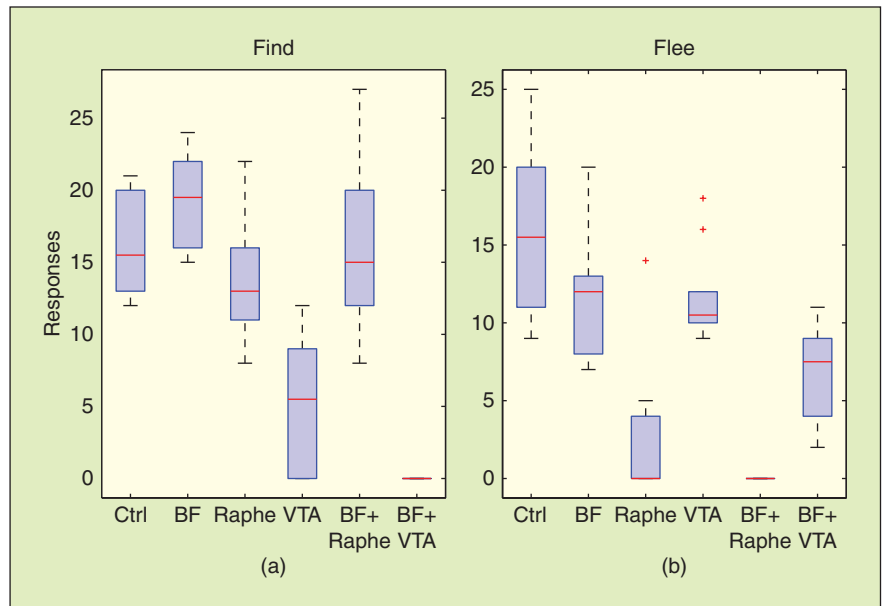


Figure 4. Behavioral responses for the ten subjects with an intact simulated nervous system (control), lesion of the simulated Raphe nucleus (Raphe), lesion of the simulated VTA, lesion of the BF, and lesions of multiple areas (BF + Raphe and BF + VTA). (a) Find responses; (b) Flee responses.

deviation. The other comparisons were not significantly different ($P > 0.01$; t -test).

The responses of the neuromodulatory systems were strongly correlated with colors that predicted value. Red predicted threatening stimuli and Raphe activity increased in the presence of red [see Figure 6(a)]. Green predicted positive valence stimuli, and the VTA activity increased in the presence of green [see Figure 6(b)]. Although neuromodulatory responses increased with colors that predicted value, their responses decreased for colors that were value-independent

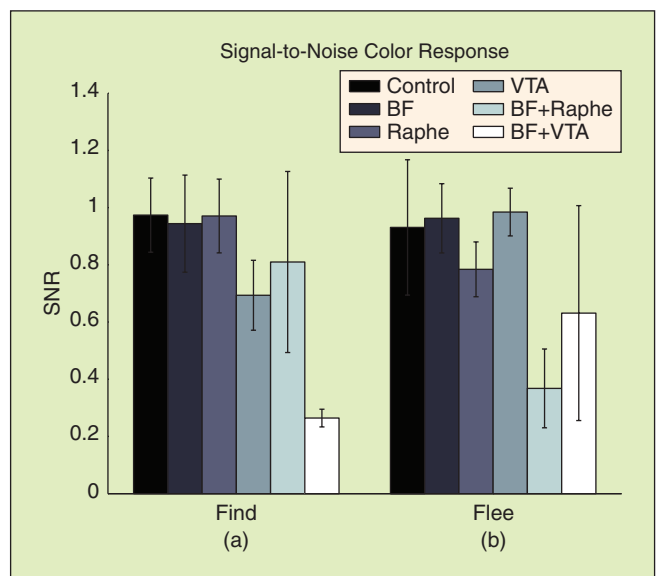


Figure 5. SNR of target color activity to all color activities during the Find and Flee behaviors. (a) Find responses. (b) Flee responses.

and that were predicted a value not associated with a particular neuromodulatory system (see Figure 6).

Discussion

In this article, we used a cognitive robot, CARL-1, to test the hypothesis that neuromodulatory activity can shape learning, drive attention, and select actions. CARL-1 learned to approach stimuli that were predictive of positive value and move away from stimuli that were predictive of negative value (see Figures 3 and 4). An intact neuromodulatory system was necessary for

correct behavioral responses (see Figure 4) and for appropriate neuromodulatory responses to stimuli (see Figures 5 and 6). These experiments suggest a mechanism of how neuromodulatory systems influence attention and decision-making.

The neural control of the cognitive robot presented here may be a design strategy for controlling autonomous systems based on the principles of neuromodulation found in the mammalian brain. Such a controller would flag an important environmental stimulus, cause the autonomous system to focus its attention on the appropriate signal, ignore irrelevant distracters, and quickly respond to pressing events.

Dopamine and Wanting Behavior

Dopamine appears to be important for wanting, i.e., the motivation process in acquiring an object [13]. Dopamine, which is found throughout the central nervous system, is produced in the VTA. A recent proposal ties the prediction error to wanting by suggesting that incentive salience is the expected future reward that maps actions to rewards [14]. Alternatively, it has been proposed that dopamine is involved with the discovery of new actions, and it influences action-outcome contingencies [11]. From the evidence, it appears that dopamine is an important signal for the acquisition of value-laden objects.

In this article, we showed that DA neuromodulation arising from a simulated VTA was necessary for value-laden wanting responses. When CARL-1's DA system was intact, it approached stimuli that were predictive of positive value and ignored neutral stimuli [see Figure 3(a)–(g) and control group in Figure 4]. When CARL-1's VTA was lesioned, the number of Find responses, which signify wanting, significantly decreased (see VTA group in Figure 4). Instead of approaching these positive-value stimuli, CARL-1 treated green objects as neutral stimuli.

Serotonin and Risky Behavior

Serotonin originates in the Raphe nucleus, and its effect on the nervous system appears to be related to the control of stress. The structures that receive serotonin from the Raphe modulate behavioral response to threats and risks [6]. For example, serotonin plays an important role in social anxiety and social threats in primates [15].

In our experiments with CARL-1, we showed that 5-HT neuromodulation arising from a simulated Raphe nucleus was needed to respond appropriately to threatening stimuli. When CARL-1's 5-HT system was intact, it moved away from threatening stimuli and ignored neutral stimuli [see Figure 3(h)–(n) and control group in Figure 4]. However, when CARL-1's Raphe was lesioned, its behavior became risky in that it approached red stimuli as if they were of neutral value (see Raphe group in Figure 4).

Acetylcholine and Attentional Effort

Acetylcholine originates from the BF and projects to the cortex, amygdala, and hippocampus. The BF appears to enhance input processing and the allocation of attentional resources for important stimuli under challenging conditions [16]. Removal of ACh projections to the parietal and frontal cortex impairs the ability to increase attentional effort [17].

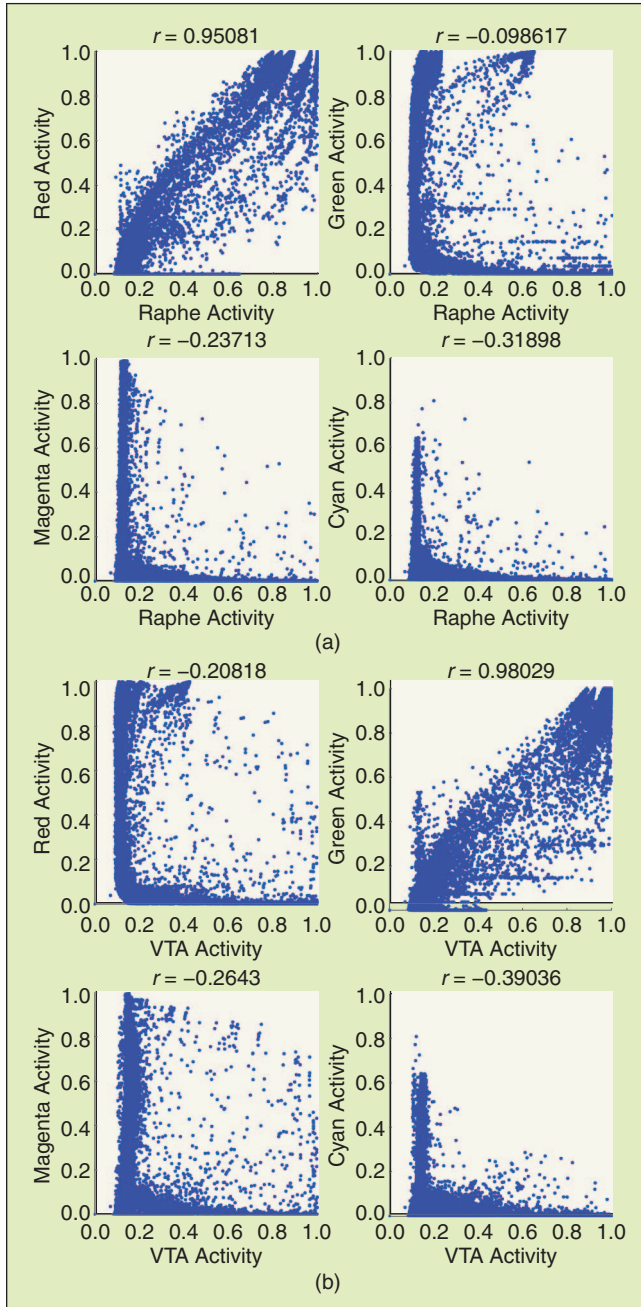


Figure 6. Scatter plots of visuomotor activity versus neuromodulatory activity for ten subjects over all control trials. The Pearson's correlation coefficient (r) is given at the top of each plot. (a) Color neural activity versus Raphe activity. (b) Color activity versus VTA activity.

In our experiments, the simulated BF enhanced CARL-1's ability to attend to salient objects. Removal of the BF alone through simulated lesions did not have a significant effect on CARL-1's behavior or the signal-to-noise response in the visuomotor area (see BF in Figures 4 and 5). However, removal of the BF and another neuromodulatory area such as VTA or Raphe significantly reduced the appropriate behavioral responses and the SNR well below the levels where only Raphe or VTA were lesioned (see BF + Raphe and BF + VTA in Figures 4 and 5). This suggests a compensatory mechanism for the BF and other regions, and it is in agreement with the notion that ACh increases the allocation of attentional resources.

Role of Phasic Neuromodulation

Phasic bursts of neuromodulatory activity were necessary to shape CARL-1's behavior during the training and drive appropriate behavioral responses during the testing. The phasic response of the simulated neuromodulators caused CARL-1 to attend to appropriate stimuli, ignore distracters, and take decisive actions (see Figure 3). Neuromodulator activity was strongly correlated with stimuli that were value laden (see Figure 6). When phasic neuromodulation was impaired, the SNR of the system decreased (see Figure 5) and CARL-1 made poor decisions (see Figure 4). This link between phasic neuromodulation and accurate action selection is in agreement with empirical data from animal models [11]. It appears that phasic neuromodulation is important for shifting attention when environmental demands require such vigilance [5].

The Neurorobot Approach

Neurorobotics and cognitive robotics are emerging fields in computer science, neuroscience, and engineering [18]. Neurorobots not only provide a tool for studying brain function by embedding neural simulations on a robotic platform but also provide the groundwork to develop intelligent machines based on neurobiological principles. The present work showed how a model of neuromodulation could be used to shape a robot's behavior, such that it focused its attention on important events and made effective decisions.

Although it could be argued that virtual environments could be used for the present work, the real environment is required for several reasons [19], [20]. First, simulating an environment can introduce unwanted and unintentional biases into the model. For example, a computer-generated object presented to a vision model has its shape and segmentation defined by the modeler and directly presented to the model. In a simulation, the color and shading of an object is typically uniform and noise free. However, a device that views objects on the floor of a room has to segment the shape and figure from the ground based on its own active vision and deal with camera sensor noise, occlusions, viewing angles, and varying light conditions. Second, because real environments are rich, multimodal, and noisy, an artificial design of such an environment is computationally intensive and difficult to simulate. However, all these interesting features of the real world come for free when the robot is allowed to freely move and actively sense in an environment. Finally, there are theoretical implications that can be

Bursts of cholinergic, dopaminergic, and serotonergic activity sharpens attention and lead toward appropriate action selection.

characterized by the slogan understanding through building [20]. To truly understand the system being studied, it is essential to build the actual physical system. Real physical systems tend to yield the most insights, because they include the most details in their design and are grounded in the physics of the real world.

In the work presented here, CARL-1 overcame much of the environmental and sensory noise through phasic neuromodulation. Because the environment was varied and interesting, it developed interesting, experience-dependent responses that would be difficult to replicate in a simulated environment. For example, the Find and Flee responses that emerged through CARL-1's learning were fairly complex. CARL-1 would focus its attention on a salient object, by aiming its camera at the object of interest, as it either approached or moved away from the stimulus (see Figure 3). In particular, the Flee response gave the impression of an animal warily eyeing a threatening object as it slowly backed away.

Neuromodulation as a Robot Controller

While conventional robots and autonomous systems require some level of supervision and tuning of parameters to fit a particular domain, biological organisms have the ability to respond quickly and appropriately in an ever-changing world. We have shown how a model of the neuromodulatory system and surrounding regions can cause a robot to sharpen its sensory systems, attend to behaviorally relevant objects and ignore distractions, learn to predict the value and outcome of its decisions, and respond decisively and appropriately to environmental events.

Other groups have taken a similar approach in modeling neuromodulation and action selection. The phasic response of dopamine has been modeled to examine reward anticipation behavior in a robot [21]. Models of the basal ganglia have been tested on robots to demonstrate action selection and switching behavior [22]. In a robotic system that has correlates with features of the noradrenergic system, cyber rodents explored new behaviors when their battery packs were full but took more exploitative behavior when their battery packs were nearly empty [23]. A study of selection and learning in a simulated robot showed how modulating attentional effort could induce learning and memory [24]. These studies, which use neural architectures to guide behavior and test models of cognition, are in a similar vein to the present work.

Our work with CARL-1 differs from the aforementioned studies in that it describes a specific neural mechanism for neuromodulation and shows how this mechanism can lead to decisive behavior under noisy conditions. That is, a neural network can quickly change from arbitrary responses to a winner-take-all response by amplifying connections carrying sensory information. This mechanism has been shown in the present experiments with CARL-1, in theoretical modeling [1] and in empirical data [3], [4].

Cognitive robots and neurorobots provide a synergy between empirical and simulated data, which can lead to improvements in the model and predictions in the modeled organism. An advantage of the neurorobot approach, taken here, is that it provides a model that can be directly tested against animal models, both in its behavioral response and neuronal response. Another advantage of this approach is that cognitively and neurally inspired robots can provide a framework for a new class of intelligent machines. We have presented a design strategy, based on the principles of the neuromodulatory system, which controlled the behavior of autonomous robot systems. This research showed that such a system could respond appropriately to environmental changes. Although the field is at a nascent stage, researchers in cognitive robotics are following working models: biological nervous systems and human cognition. If scientists are able to find the underlying principles of these working models and engineers can construct machines based on these principles, it will result in a major advancement in the field of robotics.

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Keywords

Autonomous agents, behavior-based systems, learning and adaptive systems, neurorobotics.

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